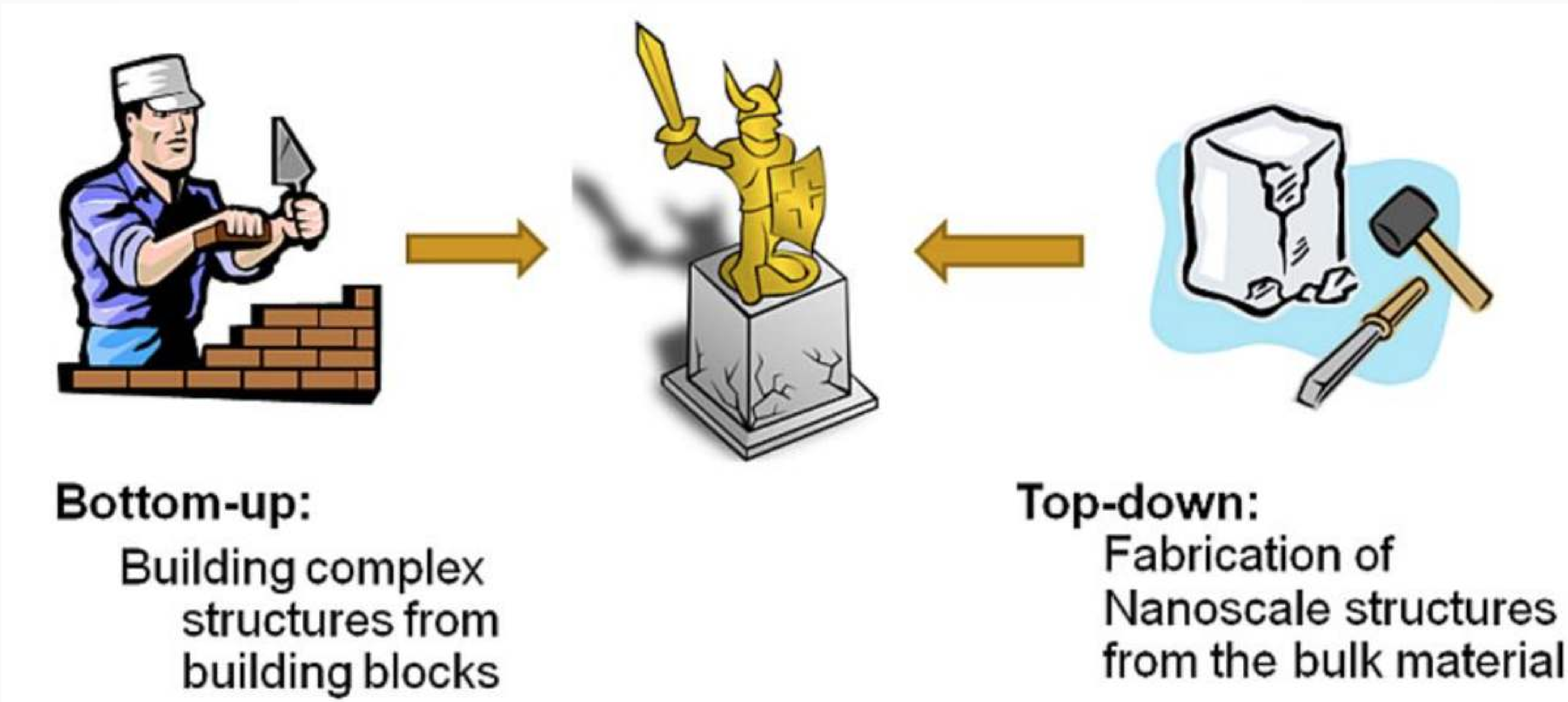
MT4013 - Material Sensor

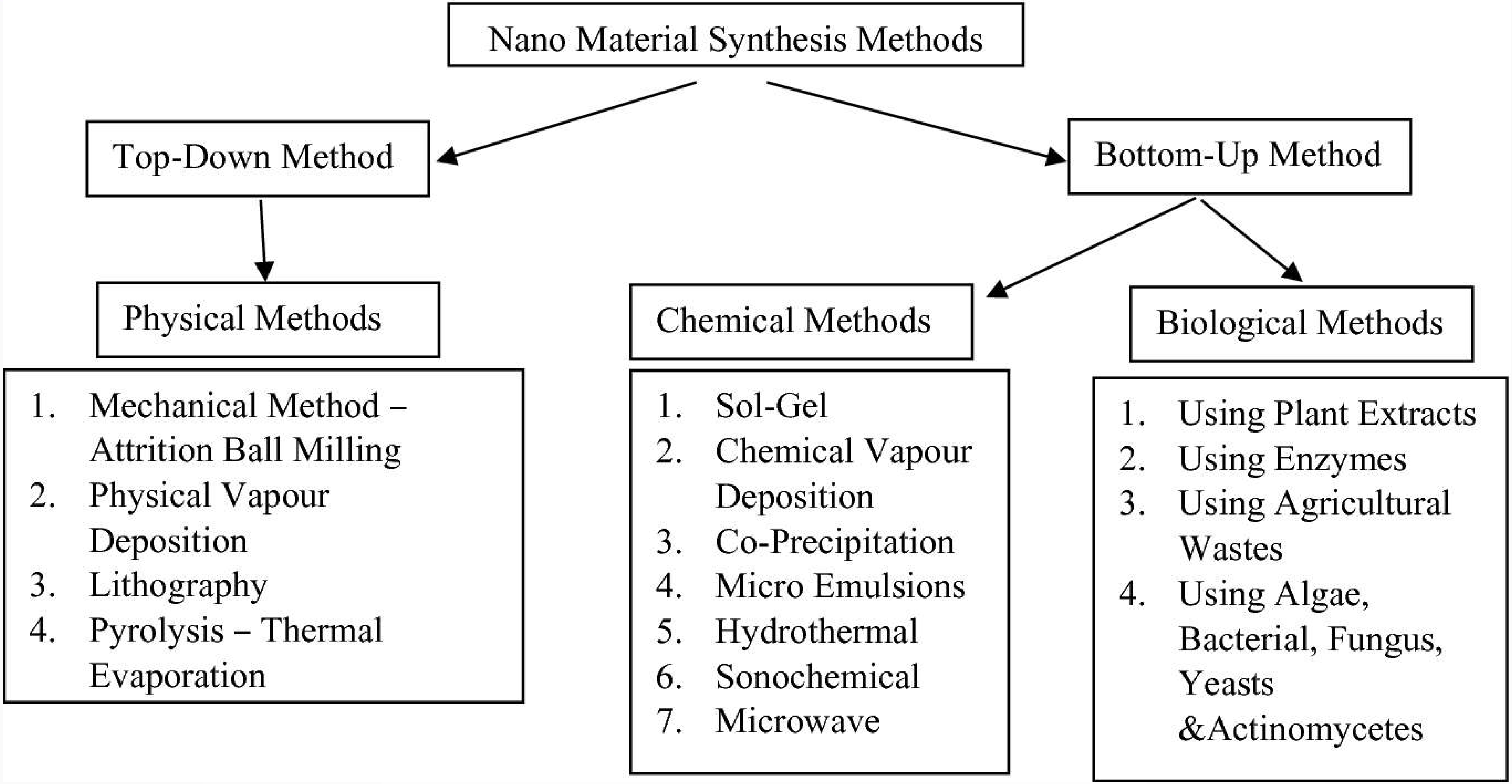
Teknik Fabrikasi Nanomaterial Rute Kimia

Eka Nurfani
Program Studi Teknik Material, Institut Teknologi Sumatera

Manufacturing of Nanomaterials

Approaches: (1) Bottom-up; (2) Top-down





Narsimha P. et al. (2020) Synthesis of Nanomaterials for Energy Generation and Storage Applications

Chemical Method

1) Sol-gel
2) Chemical Vapour Deposition
3) Micro Emulsion
4) Hydrothermal
5) Sonochemical
6) Microwave

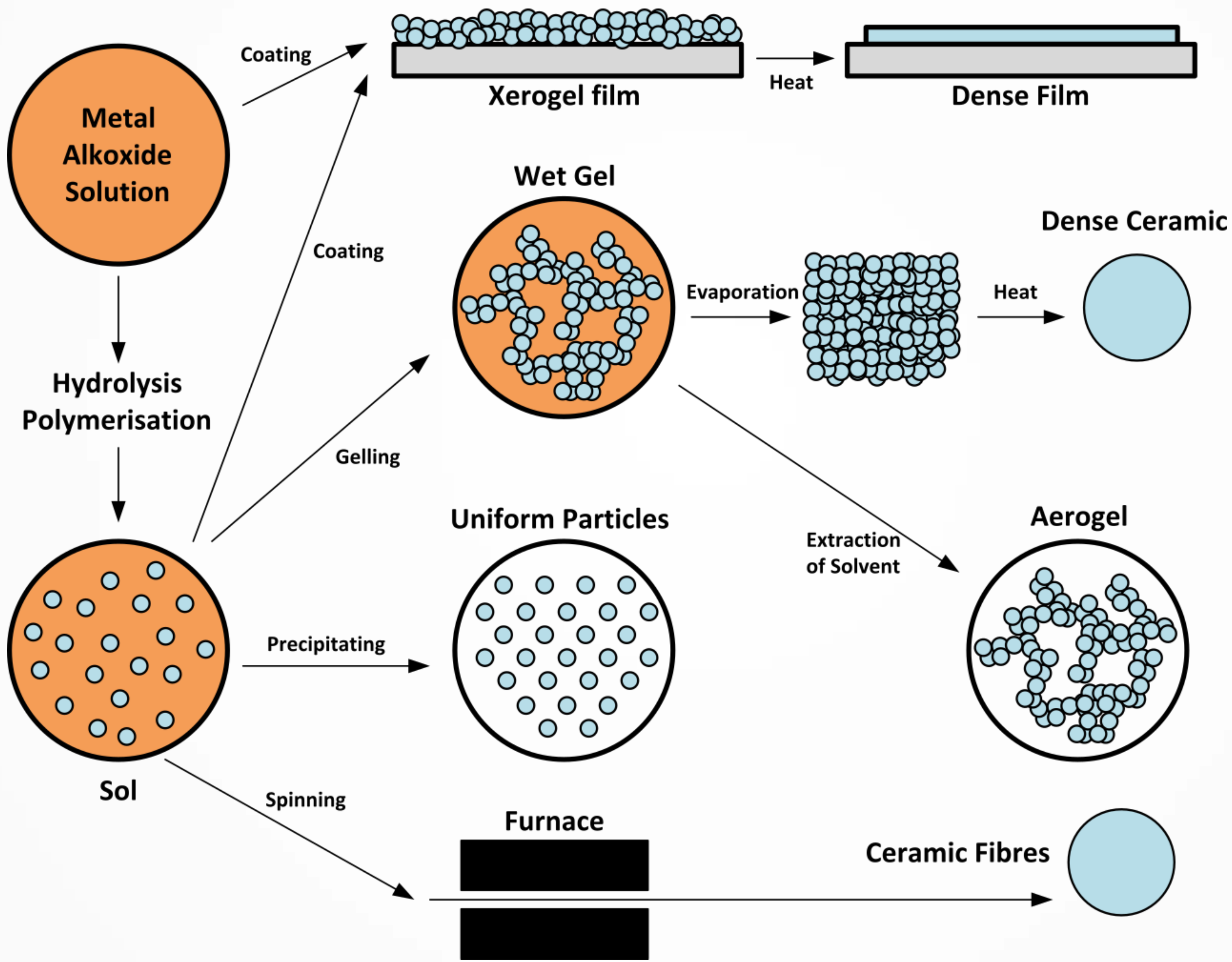
Sol-gel

Sol-gel process is a chemical route used to synthesize glassy or ceramic materials at relatively low temperatures, based on wet chemistry processing, that involves the preparation of a sol, the gelation of the sol and the removal of the liquid existing in fine interconnected channels within the gel.

Advantages:

- Versatile: better control of the structure, including porosity and particle size; possibility of incorporating nanoparticles and organic materials into sol-gel-derived oxides;
- Extended composition ranges: it allows the fabrication of any oxide composition, but also some non-oxides, hybrid organic-inorganic materials,
- Better homogeneity: due to mixing at the molecular level; high purity
- Less energy consumption: there is no need to reach the melting temperature
- Coatings and thin films, composites, porous membranes, powders and fibers;
- No need for special or expensive equipment.

Sol-gel

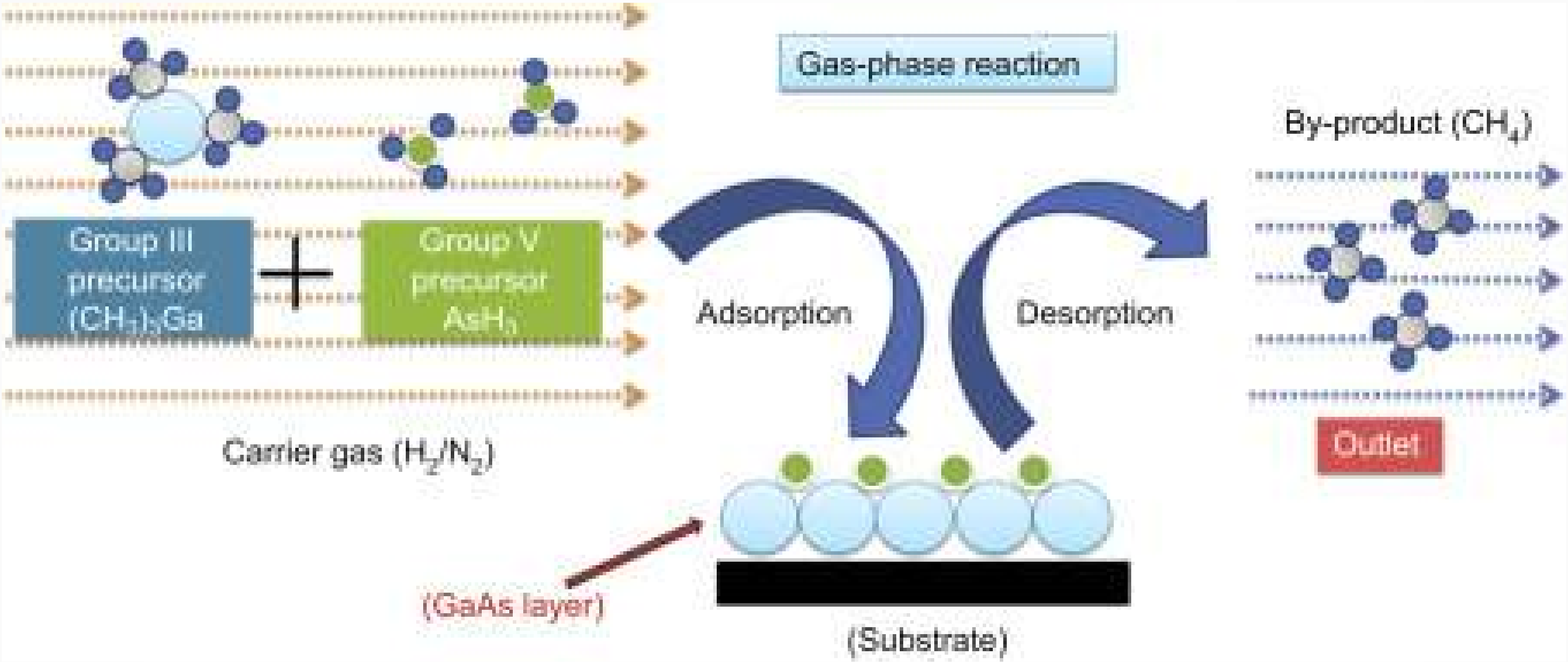


Metal Alkoxide Solution
Coating
Xerogel film
Heat
Dense Film
Coating
Wet Gel
Dense Ceramic
Evaporation
Heat
Hydrolysis Polymerisation
Gelling
Uniform Particles
Extraction of Solvent
Aerogel
Precipitating
Sol
Spinning
Furnace
Ceramic Fibres

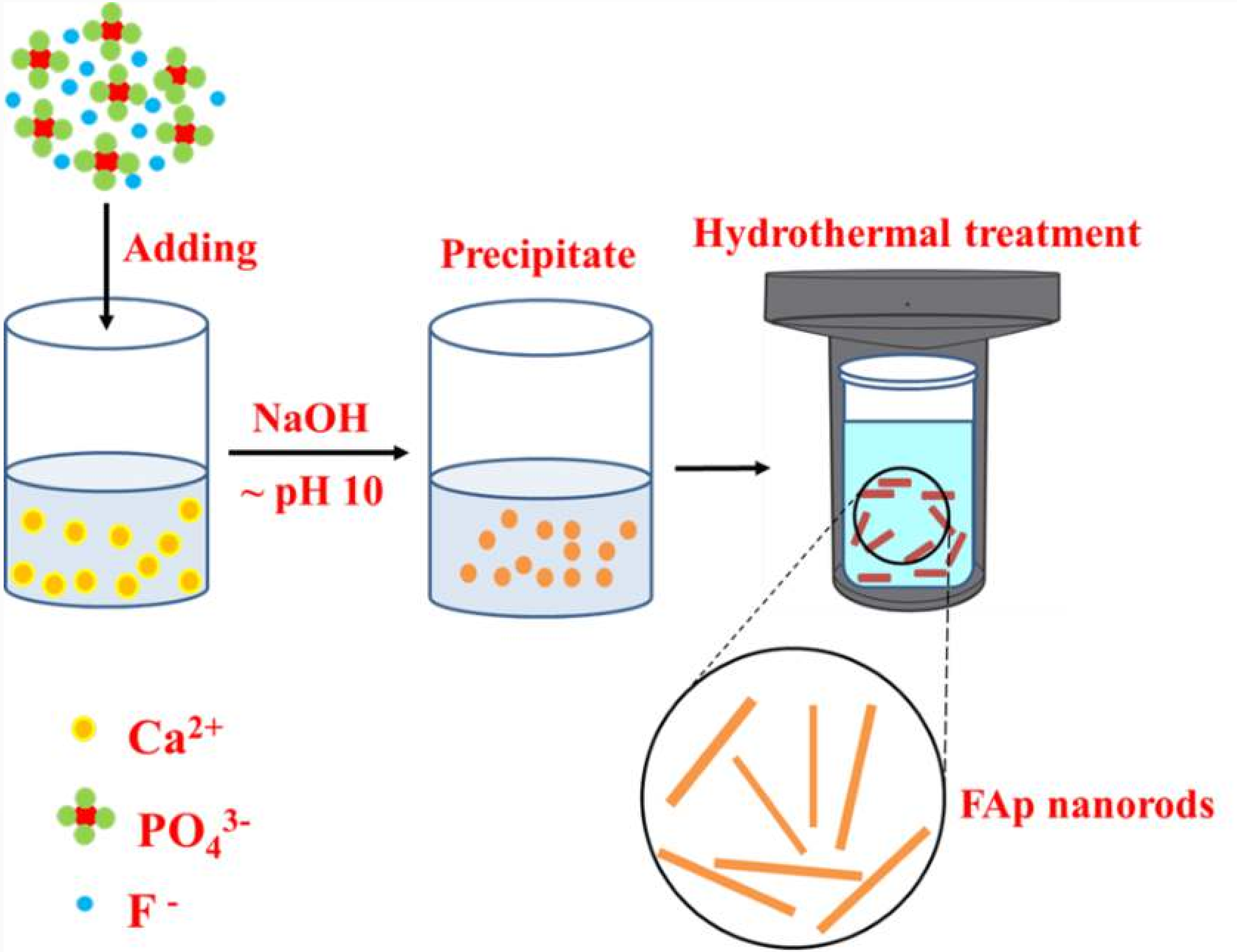
Chemical Vapour Deposition

- Metal organic CVD
- Low pressure CVD
- Athmosperic pressure CVD
- Plasma Enhanced CVD

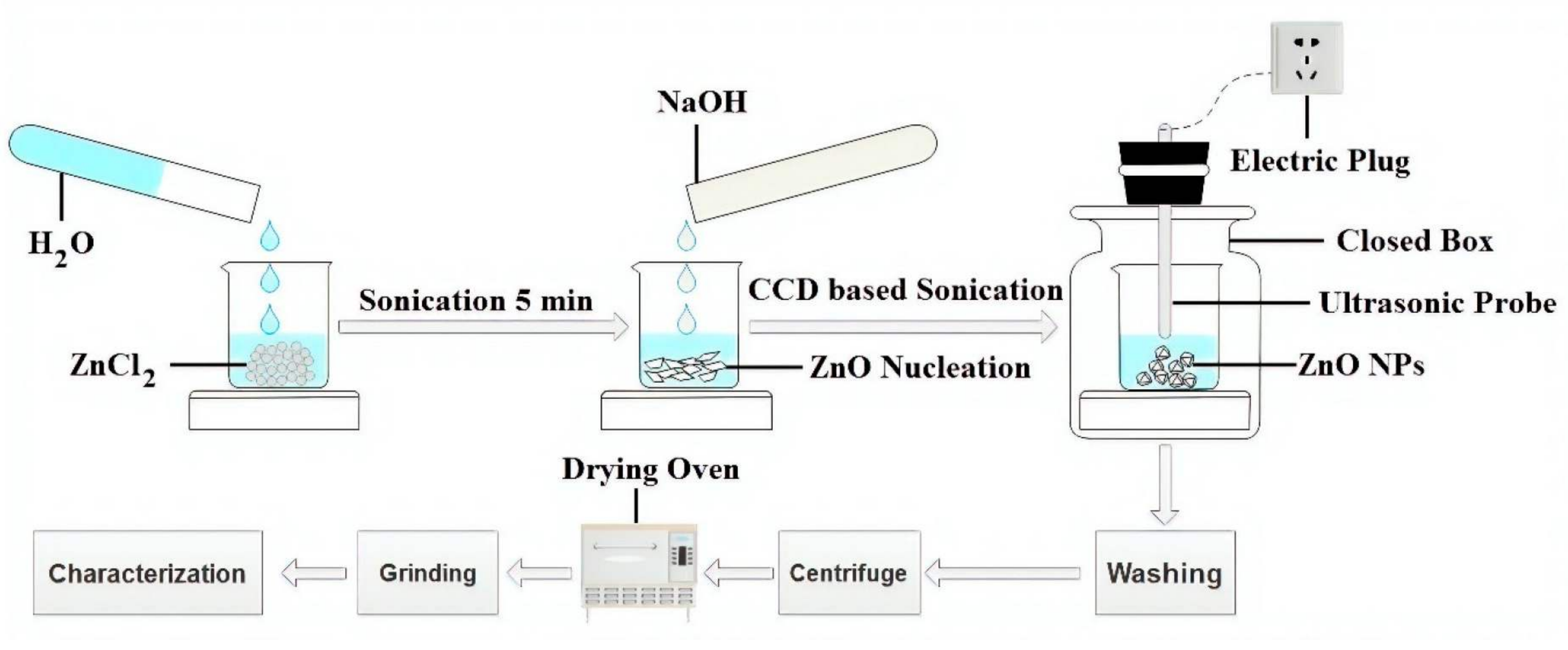
Metal organic CVD



Hydrothermal

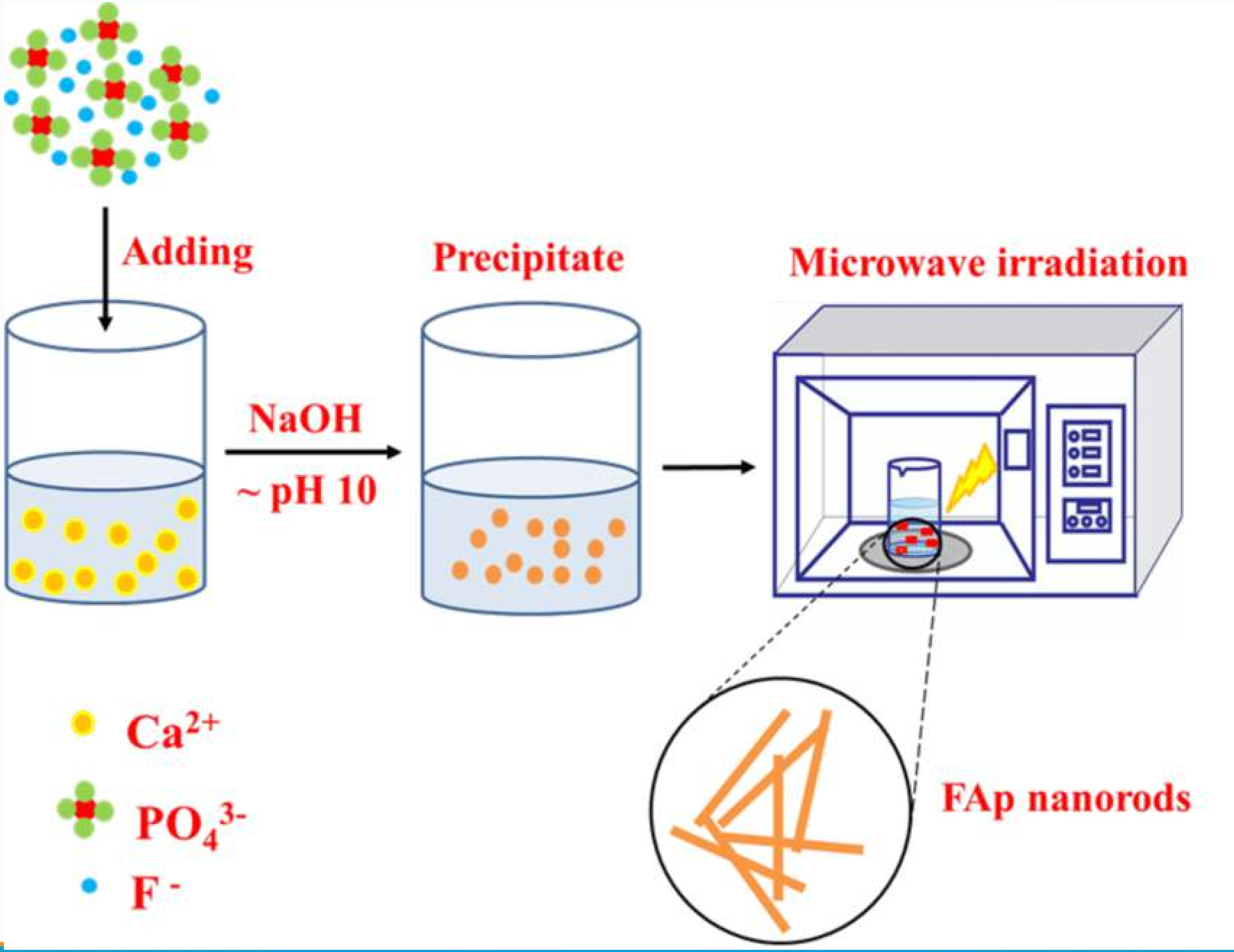


Sonochemical



Sonochemical synthesis is the process which utilizes the principles of sonochemistry to make molecules undergo a chemical reaction with the application of powerful ultrasound radiation (20 kHz - 10 MHz).

Microwave



Adding
NaOH
~ pH 10
Precipitate
Microwave irradiation
FAp nanorods
Ca^{2+}
PO_4^{3-}
F^-